What Local Costs: A Paired Evaluation of a Single-GPU Financial RAG Pipeline Against a Published API-Based System

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***Abstract*—Retrieval-augmented generation (RAG) systems for financial filings are usually built on paid APIs and evaluated only on the quality of the final written answer, which leaves two questions unanswered: how much of a system's performance comes from retrieval rather than generation, and what is lost by running the whole pipeline locally. We build FinRAG, a RAG pipeline over 497 S&P 500 10-K filings that runs entirely on one 12 GB laptop GPU with no paid services, fine-tuning a 33M-parameter embedder, a cross-encoder reranker and three generators (3B–7B) with QLoRA. We then evaluate it against HDRR [29], a published 2026 system on the same benchmark, under an unusually strict protocol: the same 258 held-out queries, HDRR's own GPT-4.1 judge, rubric and prompt verbatim, and a paired bootstrap over per-query scores. We first validate our grader by re-scoring 40 of HDRR's own published answers, reproducing them at 90% exact agreement, 100% within one point and $r = 0.993$. FinRAG scores 6.02/10 against HDRR's 7.63 (95% CI $[-2.08, -1.13]$), and is statistically indistinguishable from one of their five published configurations. Supplying gold evidence raises FinRAG to 6.70, which decomposes the gap into 0.67 attributable to retrieval and 0.93 to generator capacity. On retrieval — which HDRR does not report — FinRAG reaches NDCG@10 0.599 on filings never seen in training. Finally, we replace HDRR's per-query LLM ticker extraction with a dictionary built from public SEC company-name data: it routes 90.8% of queries at 99.3% precision and agrees with their LLM router on 100% of the 232 queries where both commit, at zero inference cost. We argue that the useful contribution of local financial RAG is not higher answer quality but a measured account of what quality costs.**

***Index Terms*—retrieval-augmented generation, dense retrieval, document routing, financial NLP, question answering, QLoRA, LLM-as-judge, reproducibility**

I. Introduction

Financial analysts answer narrow, factual questions by reading long regulatory filings: how a company's segment revenue moved between two years, what a specific accounting policy says, which risks were disclosed. Large language models are prone to fabricating such figures, and retrieval-augmented generation (RAG) [1] mitigates this by grounding the answer in retrieved passages.

Published financial RAG systems share two properties that limit what can be learned from them. First, they depend on paid APIs — typically for embeddings, for query or entity rewriting, and for generation — so their reported quality is partly a statement about the budget behind them. Second, they are evaluated end to end, usually by an LLM judge scoring the final answer. An end-to-end score cannot say whether a wrong answer came from retrieving the wrong passage or from reasoning badly over the right one, and the systems that report it generally publish no retrieval metrics at all, even when the benchmark supplies gold evidence.

This paper takes the opposite position deliberately. We build a pipeline constrained to one consumer GPU and no paid inference, measure each stage separately, and then compare the whole system against a published API-based system on that system's own terms. The comparison is a loss, and we report it as one; the contribution is in what the loss can be decomposed into.

Our contributions are:

- **A paired evaluation protocol against a published system, with grader validation.** HDRR [29] releases per-query judge scores, which allows a paired comparison on identical questions rather than a comparison of two aggregates. We restrict to queries whose filings never appear in our training data, re-use their judge, rubric and prompt verbatim, and first demonstrate that our grading harness reproduces their published scores ($r = 0.993$). We are not aware of prior financial RAG work that validates its LLM judge against a baseline's own published outputs before comparing to them.
- **A decomposition of the gap to a frontier-API system.** By evaluating the same generator on retrieved and on gold evidence, we separate the 1.61-point deficit into 0.67 from retrieval and 0.93 from generator capacity — a distinction that an end-to-end score cannot express, and which HDRR cannot compute for its own system.
- **Document routing without an LLM.** HDRR extracts a company ticker with one LLM call per query. We replace it with a dictionary derived from public SEC company-name data plus an acronym stoplist, which recovers most of that capability at zero cost and agrees with their router

on every query where both commit.

- **Retrieval metrics on FinDER under a leakage-free protocol**, including graded multi-chunk relevance labels built by a coverage criterion that is stable across chunk sizes, and a filing-level train/test split.

II. Related Work

Dense retrieval. Dense Passage Retrieval [2] established contrastively trained dual encoders over sparse baselines such as BM25 [3]. ColBERT [4] trades cost for accuracy via late interaction; Sentence-BERT [5] introduced siamese sentence embeddings. Contriever [6], E5 [7], GTR [8] and the BGE family [9] are strong general-purpose retrievers. BEIR [10] is the standard zero-shot comparison suite. We build on BGE because its small variant (33M parameters) is the largest model that leaves room for a 7B generator on a single 12 GB card.

Reranking and fusion. Cross-encoder rerankers [11] rescore a first-stage candidate list. Reciprocal Rank Fusion [12] merges dense and sparse rankings. HyDE [13] rewrites the query into a hypothetical document; we do not use it, as it requires generation at retrieval time.

Parameter-efficient fine-tuning. LoRA [14] trains low-rank updates; QLoRA [15] adds 4-bit quantization so a 7B model fits on consumer hardware. Our generators are Qwen2.5 [16], Llama-3.2 [17] and Mistral [18].

Financial NLP and benchmarks. FinBERT [19] and BloombergGPT [20] are domain-adapted language models; FinQA [21] targets numerical reasoning. FinDER [22] is the retrieval-oriented benchmark we use, built on S&P 500 10-K filings with gold evidence passages. Its own paper reports retrieval as RAGAS [23] Context Recall rather than rank-based metrics.

The systems we compare against. Cheng *et al.* [28] built a financial RAG system on FinDER with off-the-shelf embeddings, a cross-encoder reranker and a paid LLM for query rewriting and generation, evaluated with an LLM judge. Their follow-up, HDRR [29], adds hybrid document-routed retrieval, in which an LLM first extracts the target company and retrieval is then confined to that company's filing. HDRR is our primary comparison point: it is recent, it uses the same benchmark, and — critically — it publishes per-query judge scores, which makes a paired comparison possible. Neither paper reports retrieval metrics.

LLM-as-judge. Automatic judging with a strong model [27] is now standard for open-ended generation, with known caveats including position bias and self-preference. We address the most relevant one here — that the judge and the baseline's generator come from the same model family — in Section VIII.

III. Dataset and Evidence Alignment

We use FinDER [22]: 497 S&P 500 10-K filings and 5,703 questions, each with a gold answer and one or more *reference* passages quoted from the filing. Questions are short and heavily abbreviated (e.g. "Delta in CBOE Data & Access Solutions rev from 2021-23"), and a substantial minority require a small arithmetic step.

TABLE I
FinDER as processed in this work.

Property	Value
Filings (10-K)	497
Questions	5,703
Chunks (448 tokens, 112 overlap)	181,867
Questions with induced labels	5,303 (93.0%)
Relevant chunks per question (mean / median)	2.18 / 2
References reaching 80% coverage	99.2%
Train / test filings	392 / 98
Train / test questions	4,215 / 1,088

Chunking. We parse each filing to text and split it into chunks of 448 tokens with 112 tokens of overlap, measured *in the retriever's own tokenizer* rather than in words. This matters: a chunk defined in words can exceed the encoder's 512-token limit and be silently truncated at encoding time. The result is 181,867 chunks.

Graded, filing-restricted relevance labels. FinDER gives evidence as raw text, not chunk identifiers, so relevance labels must be induced. A single-best-match criterion is both too coarse (evidence spans several chunks; mean 2.18 in our labels) and too permissive (10-K boilerplate is near-identical across companies, so a global match frequently lands in the wrong filing). We therefore resolve each question to exactly one filing first, by summed TF–IDF similarity over its references, and then select chunks *within that filing* by greedy union coverage: chunks are added until their union covers 80% of the reference's token multiset. The chunk contributing the largest share is graded 2 (primary evidence), others 1 (supporting).

Union coverage is a property of the evidence rather than of the chunk size, so the threshold does not need re-tuning when chunking changes — unlike a per-chunk overlap threshold, which must be lowered as chunks shrink. This yields labels for 5,303 of 5,703 questions (93.0%), with 99.2% of references reaching the coverage target and, by construction, no question whose gold spans more than one filing. Table I summarizes the corpus.

Leakage-free splitting. We split at the *filing* level, not the question level: 392 filings (4,215 questions) for training and 98 filings (1,088 questions) for evaluation. Because many FinDER questions target the same filing, a question-level split would let a retriever memorize the exact passages it is later tested on. Retrieval at inference still searches the full 181,867-chunk corpus; only training is restricted.

Label audit. Our filing-resolution step includes an override that trusts an explicitly named ticker over the TF–IDF result. Ticker strings collide with financial vocabulary — `IP` is also intellectual property, `CF` cash flow, `NI` net income — so we audited the labels for questions matching that failure mode exactly. On the test split, zero questions are affected. On the training split, 166 (3.9%) match the pattern, but their selected gold chunks cover the query text as well as any other

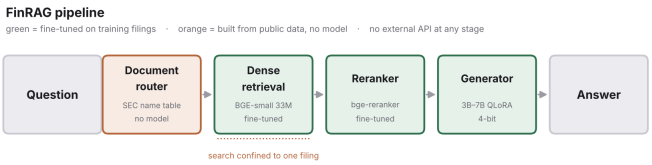


Fig. 1. The FinRAG pipeline. Shaded stages are fine-tuned on FinDER's training filings; the router is constructed from public SEC data and is not learned.

question's (0.477 versus 0.450), indicating genuine mentions of those companies rather than mislabelling.

IV. Method

Fig. 1 shows the pipeline. Four components are trained or constructed by us; none of them call an external service.

A. Embedding fine-tuning

The first-stage retriever is `bge-small-en-v1.5` [9] (33M parameters), fine-tuned with Multiple Negatives Ranking Loss. For a query q_i with positive p_i^+, cosine similarity $s(\cdot,\cdot)$ and temperature τ,

$$\mathcal{L}_i = -\log \frac{\exp\big(s(q_i, p_i^+)/\tau\big)}{\sum_j \exp\big(s(q_i, p_j)/\tau\big)}. \tag{1}$$

Two details materially affected results. Positives are drawn from grade-2 chunks rather than arbitrarily, since an arbitrary choice frequently selects a supporting fragment instead of the passage that actually answers the question. Negatives are drawn under an exclusion band: labelled gold, the immediate neighbours of gold chunks (which share overlap tokens by construction), and any candidate whose content overlap with the gold text exceeds 0.35 are all excluded, because training the model to push away text it should retrieve is actively harmful. We mine four in-filing and four cross-filing negatives per query; in-filing negatives teach fine-grained relevance, cross-filing negatives teach company discrimination.

B. Reranker fine-tuning

The reranker is `bge-reranker-base`, trained on (query, positive, negative) pairs with binary cross-entropy. Negatives must be mined from the *fine-tuned* retriever: the reranker sees that retriever's candidates at inference, and training on easier base-model negatives leaves it unprepared. Passages are clipped in the reranker's own tokenizer (XLM-RoBERTa) to 460 tokens, matching training exactly — our 448-BGE-token chunks measure a mean of 475 tokens under that tokenizer, so roughly 15% would otherwise lose their tail at inference.

C. Document routing without an LLM

Only about 11% of the fine-tuned retriever's top-100 come from the question's own filing: 10-K filings are structurally near-identical, so most retrieved text is simply the wrong company. Restricting the candidate pool to one filing is therefore the single largest available improvement, and HDRR reaches the same conclusion, implementing it with one LLM call per query to extract the ticker.

We implement it without a model. A first version matched capitalized tokens against corpus tickers and routed when exactly one matched, which covers 78.4% of queries. It fails three ways: questions naming a company rather than a ticker ("Div policy of Intuit…"); collisions in which a financial acronym is also a ticker, so a query that *did* name its ticker is discarded as ambiguous ("Growth acceleration in SNPS Design IP segment" matches both SNPS and International Paper); and single-letter tickers, which a 2–6 character pattern cannot admit without catastrophic false positives.

Our router resolves these with a dictionary built from SEC's public `company_tickers.json`, restricted to the 497 corpus filings and normalized into name variants, plus a stoplist of financial acronyms that are also tickers. Evidence is ranked: an unambiguous ticker token wins outright, because on this corpus a unique ticker names the correct filing 100% of the time, against 97.4% for a name match; a company name is used only when no unambiguous ticker is present. Where neither resolves, the query stays unrouted and falls back to full-corpus retrieval, which is deliberately conservative — a wrong route confines retrieval to the wrong company with no escape.

D. Generator fine-tuning

We fine-tune three instruction-tuned generators with QLoRA [15] — Qwen2.5-3B, Llama-3.2-3B and Mistral-7B-Instruct-v0.3 — loading base weights in 4-bit NF4 and training rank-16 adapters on attention and MLP projections. Prompt tokens are masked so the loss is computed only on the answer. Training examples pair a question with its gold context; 6% are abstention examples with deliberately varied refusal phrasings, since a generator trained only on answerable questions will confabulate when retrieval fails.

V. Evaluation Protocol

Retrieval is scored with MRR@10, NDCG@10/@20 and Recall@k against the graded labels of Section III, using linear gain $g/\log_2(r+1)$. Generation is scored with ROUGE-L [24], BLEU [25], BERTScore-F1 [26], and a rounding-aware numeric exact-match on the subset of questions whose gold answer contains a verifiable figure. All confidence intervals come from a paired bootstrap with resampling indices shared across systems, so that per-system intervals and pairwise differences are drawn from the same resamples; independent per-system bootstraps ignore that every system is scored on the same queries and overstate uncertainty on a difference.

A. Comparison against HDRR

HDRR's repository publishes per-query judge scores for every system in their paper, keyed on FinDER question identifiers. This permits a comparison far stronger than placing two aggregate numbers side by side, provided three conditions hold.

TABLE II
RETRIEVAL ON 1,088 HELD-OUT QUESTIONS FROM 98 FILINGS NEVER SEEN IN TRAINING. RETRIEVAL SEARCHES THE FULL 181,867-CHUNK CORPUS. FT = FINE-TUNED.

System	NDCG@10	MRR@10	R@10	R@100
BM25	0.0359	0.0403	0.0613	0.1742
BGE-base, off the shelf	0.0681	0.0727	0.1030	0.2304
BGE-small, off the shelf	0.0777	0.0807	0.1134	0.2256
FT dense	0.1866	0.1994	0.2420	0.4032
FT dense + RRF	0.1678	0.1739	0.2451	0.4033
FT dense + ticker route	0.5342	0.5354	0.7021	0.8801
+ FT reranking	0.5414	0.5550	0.6706	0.8368
+ SEC-name route	**.5994**	**.6171**	**.7322**	**.8749**

No leakage. Their query splits draw from all 497 filings, 392 of which trained our models. We therefore restrict to the intersection of their published splits with our 98 held-out filings: 258 questions with recoverable ground truth. Across all seven of their systems, scores on this subset fall within ± 0.3 of their scores on their full query set, so the subset is not an unrepresentative slice.

Identical grading. We reproduce their harness exactly: model `gpt-4.1`, temperature 0, 100 max tokens, JSON-constrained output, their `BENCHMARK_EVALUATION` system and user prompts verbatim, and their score parser. Ground truth is read from their own result files rather than re-derived from the dataset, and we replicate their text sanitizer, so the judge receives byte-identical references to the ones their judge received.

Validated grader. Before comparing, we re-scored 40 of HDRR's own published answers with our harness. Agreement with their published scores was 90.0% exact, 100% within one point, mean difference $+0.05$, $r = 0.993$. This establishes that any difference we report is a difference between pipelines and not between grading setups, and it bounds judge drift between their evaluation and ours.

VI. RESULTS

A. Retrieval

Table II reports retrieval on the 1,088 held-out questions. Fine-tuning the 33M embedder more than doubles NDCG@10 over the strongest off-the-shelf baseline (0.1866 versus 0.0777, $+0.1091$, CI $[+0.0903, +0.1287]$). Document routing is the dominant single factor, $+0.3468$ (CI $[+0.3269, +0.3691]$), and replacing the ticker regex with the SEC dictionary router adds a further $+0.0580$ (CI $[+0.0460, +0.0705]$). Fig. 2 shows the progression. The complete pipeline reaches NDCG@10 0.5994, a $16\times$ improvement over BM25 and $3.2\times$ over the fine-tuned dense retriever alone.

Two negative results are worth recording. Sparse fusion *hurts* once the embedder is domain-adapted (-0.0189, CI $[-0.0314, -0.0056]$). And reranking applied to unrouted candidates is sharply harmful (-0.1021, CI $[-0.1211, -0.0841]$), while the same reranker applied *after* routing is mildly positive but not significant ($+0.0075$, CI $[-0.0156, +0.0297]$). A reranker trained on one candidate distribution and deployed on another mis-ranks; the component is not independently useful, only useful in position.

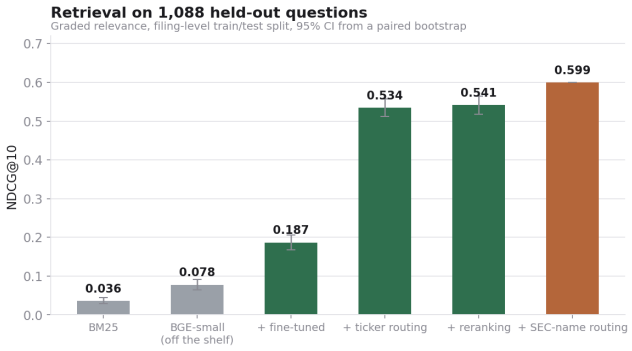


Fig. 2. Retrieval on 1,088 held-out questions. Error bars are 95% paired bootstrap intervals. Document routing, not reranking, is the dominant factor.

TABLE III
DOCUMENT ROUTING ON THE 259 SHARED QUERIES. "CORRECT" IS THE SHARE OF ALL QUERIES ROUTED TO THE FILING CONTAINING THE GOLD EVIDENCE.

Router	Cost/query	Routed	Prec.	Correct
HDRR, LLM extraction	1 call	96.1%	98.8%	**95.0%**
Ours, ticker regex	none	78.0%	100.0%	78.0%
Ours, SEC names	**none**	90.3%	98.7%	89.2%

B. Document routing

Table III and Fig. 3 compare the three routers on the 259 queries shared with HDRR, scored against the filing our labels place the gold chunks in. The SEC dictionary recovers two thirds of the gap between the regex and HDRR's LLM router at no inference cost. More tellingly, on the 232 queries where both our router and theirs commit to a filing, they select the *same* filing 232 times — the two methods are not approximating each other, they agree exactly. The residual difference is coverage, not correctness: HDRR commits on 17 queries where we abstain.

C. Generation

Table IV reports the three fine-tuned generators on the held-out questions, under retrieved context and under gold context. QLoRA raises ROUGE-L by 20–24% relative over the corresponding base model for every backbone. Mistral-7B is strongest on surface overlap; Qwen2.5-3B is strongest on numeric exact-match despite being less than half the size, which suggests numeric fidelity and fluency are not the same capability.

D. Comparison with HDRR

Table V and Fig. 4 report the paired comparison under the protocol of Section V-A. FinRAG scores 6.02 against HDRR's 7.63. The difference is real and we report it as such: the 95% CI is $[-2.08, -1.13]$ and excludes zero. FinRAG is statistically

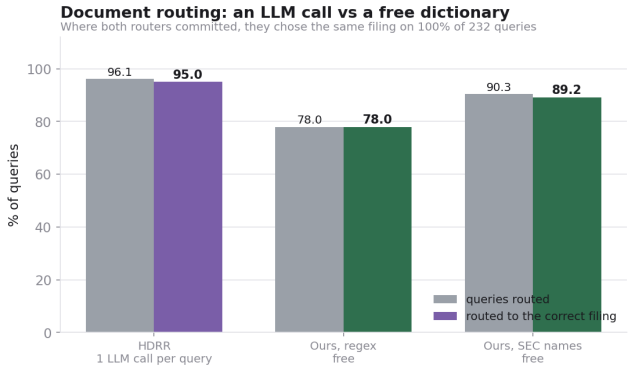


Fig. 3. Document routing: one LLM call per query against a free dictionary lookup. Where both routers commit to a filing, they choose the same one on 100% of 232 queries.

TABLE IV
GENERATION ON HELD-OUT QUESTIONS. NUMERIC EM IS COMPUTED ON THE TRUSTED NUMERIC SUBSET ($n = 121$) WITH A ROUNDING-AWARE CRITERION.

Generator	Context	ROUGE-L	BERTScore	Num. EM
Qwen2.5-3B	retrieved	0.2041	0.8626	0.107
Llama-3.2-3B	retrieved	0.2119	0.8644	0.083
Mistral-7B	retrieved	**0.2318**	**0.8712**	0.099
Qwen2.5-3B	gold	0.2229	0.8704	**0.190**
Llama-3.2-3B	gold	0.2312	0.8712	0.091
Mistral-7B	gold	**0.2518**	**0.8783**	0.140

TABLE V
PAIRED COMPARISON ON 258 SHARED HELD-OUT QUERIES, SCORED BY HDRR'S OWN GPT-4.1 JUDGE WITH THEIR RUBRIC AND PROMPT. CIS FROM 10,000 PAIRED BOOTSTRAP RESAMPLES. Δ IS OURS MINUS THEIRS; W/L/T COUNTS QUERIES WHERE OUR SCORE WAS STRICTLY HIGHER, STRICTLY LOWER, OR EQUAL.

System	Mean	Δ	95% CI	W/L/T
HDRR	7.63	-1.61	$[-2.08, -1.13]$	58/145/55
CBR + LLM route	7.44	-1.41	$[-1.91, -0.92]$	69/141/48
CBR + metadata	6.84	-0.82	$[-1.32, -0.32]$	75/133/50
Unnamed base	6.61	-0.59	$[-1.04, -0.12]$	102/105/51
Agentic RAG	6.27	-0.25	$[-0.78, +0.30]$	101/118/39
FinRAG (ours)	**6.02**	—	—	—

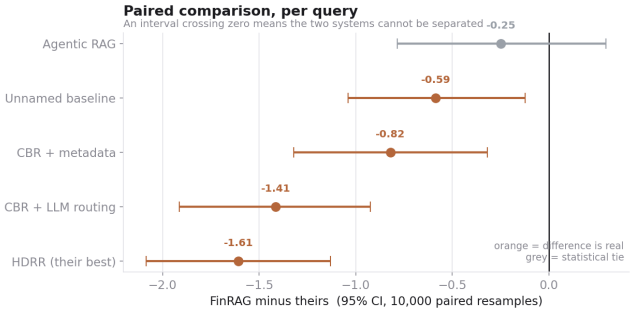


Fig. 4. Paired per-query differences against each published system, with 95% bootstrap intervals. An interval crossing zero means the two systems cannot be separated on these queries.

indistinguishable from their Agentic RAG configuration and loses to the remaining three.

E. Where the gap comes from

The value of measuring both stages is that the deficit can be attributed. Supplying the same generator with gold evidence instead of retrieved context raises its judged score from 6.02 to 6.70 (Fig. 5). The 1.61-point gap therefore decomposes as

$$\underbrace{0.67}_{\text{retrieval}} + \underbrace{0.93}_{\text{generator}} = 1.61, \tag{2}$$

and the two halves have different character. The retrieval component is a property of our pipeline and is addressable: 37% of queries receive no gold chunk in their top-3 context, and deepening the context window from 3 to 10 passages recovers part of it. The generator component is a property of the hardware budget: HDRR's answers are produced by a frontier API model, ours by a 4-bit 7B model on a laptop. At the retrieval ceiling, FinRAG is statistically indistinguishable from three of HDRR's five published configurations.

A further observation supports this reading. HDRR's scores are essentially flat with respect to *our* retrieval outcome — 7.54 on queries where our retriever found gold, 7.81 where it did not. The queries we fail are not intrinsically hard; a sufficiently strong generator compensates for weaker evidence. This is also a caution about the metric: an end-to-end judged score is relatively insensitive to retrieval quality, which is precisely why retrieval should be reported separately.

F. Efficiency

Table VI reports the deployed cost. The entire pipeline fits in 5.3–7.3 GB of VRAM. Retrieval and routing together account for approximately 0.5 s of a 12–15 s query; generation dominates. Notably, vocabulary size rather than parameter count drives memory and speed: Qwen2.5-3B, with a 152k vocabulary, is both heavier and slower than Mistral-7B with a 33k vocabulary.

VII. DISCUSSION

Routing dominates, and it does not need a model. The largest single retrieval gain in this work, +0.3468 NDCG@10, comes from confining search to one filing. HDRR reaches the same architectural conclusion and pays an LLM call per query for it. Our dictionary router reaches 89.2% correct routing against their 95.0%, and agrees with their decision on 100% of queries where both commit. For a corpus with a closed, enumerable entity set — which describes most regulatory filing collections — entity routing appears to be a lookup problem that has been treated as an inference problem.

Component quality is positional, not intrinsic. Our reranker degrades NDCG@10 by 0.10 when applied to unrouted candidates and helps marginally when applied after

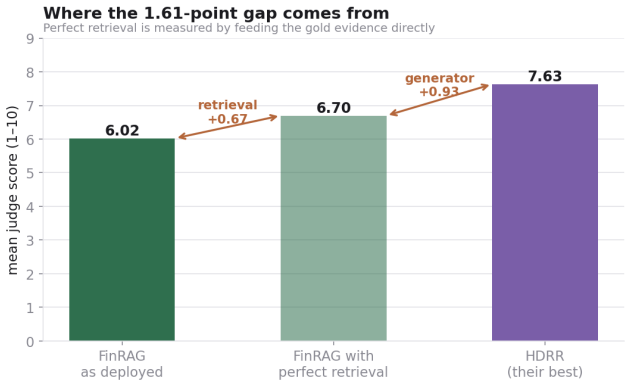


Fig. 5. Decomposition of the 1.61-point deficit. The middle bar supplies the same generator with gold evidence, isolating retrieval loss from generator capacity.

TABLE VI
DEPLOYED COST ON ONE RTX 4080 LAPTOP GPU (12.9 GB), 50 QUERIES, FULL PIPELINE CO-RESIDENT. NO EXTERNAL API CALLS AT ANY STAGE.

Generator	Peak VRAM	Tokens/s	Median latency
Llama-3.2-3B	5.32 GB	13.2	12.0 s
Qwen2.5-3B	6.30 GB	9.6	15.0 s
Mistral-7B	7.27 GB	11.4	14.3 s

routing. Sparse fusion helps a generic embedder and hurts a domain-adapted one. Neither component has a fixed value; both are functions of the candidate distribution they see. Reporting a reranker's contribution without specifying what precedes it is therefore not very informative.

Abstention is penalised by construction. HDRR's rubric instructs the judge to score 1–3 when an answer states the information could not be found, regardless of whether refusing was correct. Our generator is deliberately calibrated to abstain, and does so on 6.6% of queries; excluding those raises our mean from 6.02 to 6.34. We report the penalised figure as the headline because it is the one measured under their protocol, but note that a rubric which punishes calibrated refusal as harshly as confident fabrication rewards a behaviour most financial deployments would not want. Supporting evidence: our abstention rate falls from 6.6% to 4.3% when gold evidence is supplied, so the model abstains because context lacks the answer, not reflexively.

What local actually costs. On this benchmark the answer is 1.61 judged points, of which 0.93 is attributable to generator capacity. In exchange, the system runs in 7.3 GB on a consumer laptop, makes no external calls, and sends no filing text to a third party — which, for regulated financial data, is frequently a requirement rather than a preference.

VIII. LIMITATIONS

Our grader validation used 40 of HDRR's answers, which are predominantly high-scoring (mean 7.95); it therefore establishes agreement in the upper range more firmly than near our own operating point of 6.0. Relatedly, the judge (GPT-4.1) and HDRR's generator come from the same model family, so self-preference cannot be excluded as a contributor to the gap; we use their judge deliberately, since a different judge would make the comparison incommensurable with their published numbers, but the direction of any such bias is unfavourable to us.

Their published scores were produced several months before ours. Our replication bounds drift at $+0.05$ on 40 queries, but the two score sets are not from a single judging run.

Our relevance labels are induced rather than human-annotated. The coverage criterion is conservative and will miss chunks that contain an answer without reproducing the reference wording; an observed consequence is that queries whose labelled gold was never retrieved still improved when given more context, implying our NDCG understates true retrieval quality to some degree.

Finally, we compare against HDRR on judged answer quality only. Their paper reports no retrieval metrics, and FinDER's own paper reports retrieval as RAGAS Context Recall rather than rank-based measures, so our NDCG figures have no published comparison point on this benchmark. Establishing one is the obvious next step.

IX. CONCLUSION

We built a financial RAG pipeline that runs entirely on a single 12 GB consumer GPU with no paid inference, and evaluated it against a published API-based system under that system's own judge, rubric and prompt, after demonstrating that our grading harness reproduces their published scores at $r = 0.993$. The local system scores 6.02 against 7.63 — a real and reported loss — but the comparison decomposes it into 0.67 points of retrieval deficit and 0.93 points of generator capacity, and shows that at the retrieval ceiling the local pipeline is statistically indistinguishable from three of five published configurations. Along the way we find that document routing, the largest single retrieval factor, can be done by dictionary lookup rather than by an LLM call, at exact agreement with the learned router wherever both commit.

We suggest that the useful question for locally-hosted RAG is not whether it beats a frontier-API system, but what the difference costs and where it lives. Answering that requires reporting retrieval and generation separately, which end-to-end judged evaluation does not do. Code, labels, run files and the evaluation harness are available.[1]

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